Recap

Resolvers communicate between each other by exchanging messages

Query messages are sent by stub resolvers or resolvers (acting as clients)

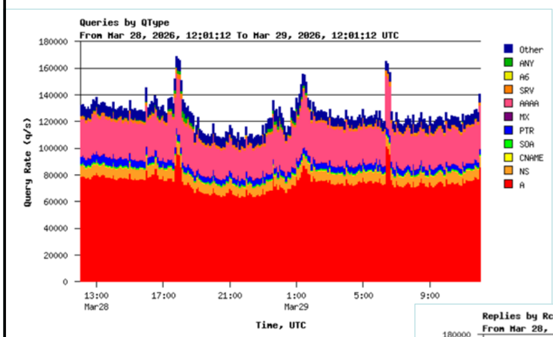
Response messages are sent by resolvers (acting as servers)

Query and response messages share the same format

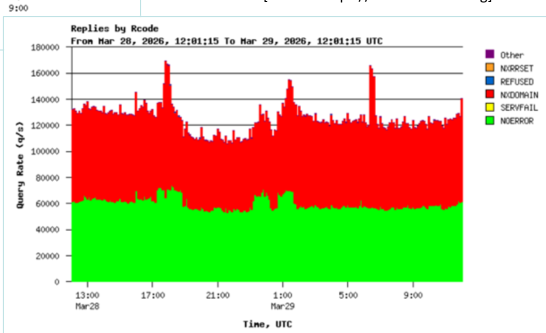
Queries specify the type of resource record being requested

Responses contain the requested records (and additional records)

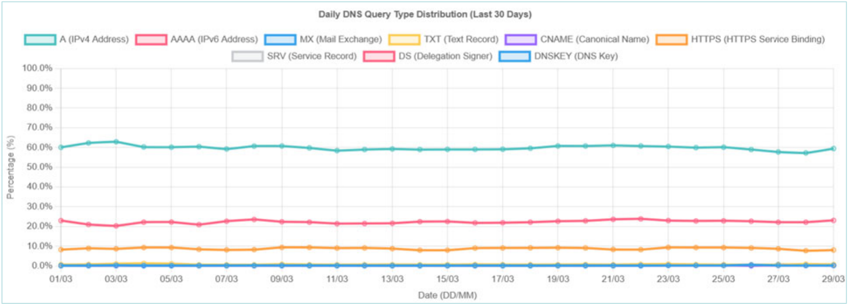
Queries/responses at `k.root-servers.org`



[Source: https://k.root-servers.org]



Cloudflare query types



131	673096.5B	61.5%
Countries	Total Queries	A (Most Common)

Country ↕	Total % ↕	A ↕	AAAA ↕	CNAME ↕	MX ↕	NS ↕	PTR ↕	HTTPS ↕	SRV ↕	DS ↕	DNSKEY ↕	TXT ↕
WORLD TOTAL	100.00%	61.47%	22.38%	0.30%	0.26%	1.27%	3.93%	9.18%	0.29%	0.26%	0.05%	0.63%
Italy (IT)	1.04%	59.12%	22.01%	0.10%	0.07%	0.34%	4.67%	12.72%	0.19%	0.18%	0.03%	0.56%

Caching

Caching mechanisms are aimed at reducing the time spent in the lookup phase and the load of the Name Servers

Caches are associated with full resolvers as well as with stub resolvers (shared vs private)

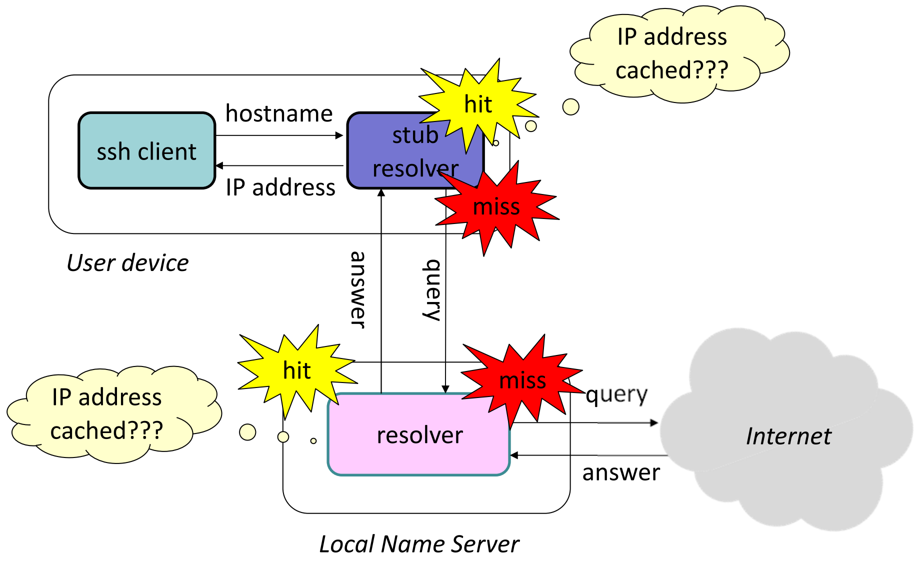
A DNS cache stores all RRs received as answers to queries (even negative answers) for possible reuse by the same query or by different queries (e.g., records about Name Servers)

Caching effectiveness is measured in terms of cache hits and misses

Very few changes to TLD Name Servers and frequent visits to popular sites/services make caching very effective

Speedup of repeated queries

Cache hits & misses



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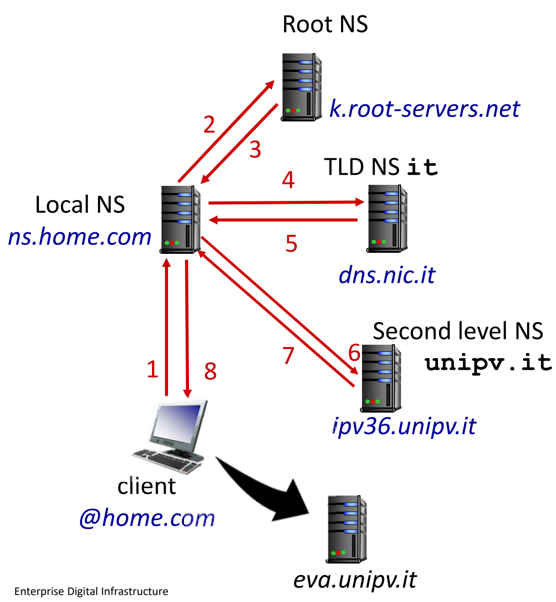
Example revisited

Client connected to the Wi-Fi *@home.com* opens a terminal to establish a secure connection to a remote machine:
`ssh eva.unipv.it`

The stub resolver queries the Local Name Server
`eva.unipv.it IN A`

The resolver of the Local Name Server queries the resolvers of the Name Servers of the hierarchy (the query does not change)
`eva.unipv.it IN A`



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Enterprise Digital Infrastructure

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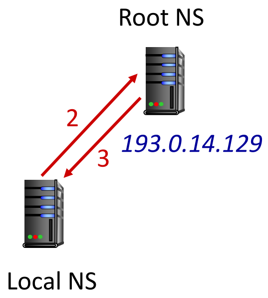
Content of local NS cache: first query



Name	Class	Type	Value	TTL
dns.nic.it	IN	A	192.12.192.5	10613
dns.nic.it	IN	AAAA	2a00:d40:1:1::5	10800
…	…	…	…	…
it	IN	NS	dns.nic.it	10720
it	IN	NS	a.dns.it	10720
…	…	..	…	…

ALL Name Servers of `it`

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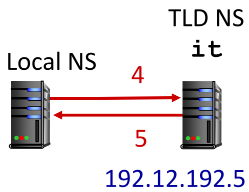
Cache content: second query



Name	Class	Type	Value	TTL
it	IN	NS	dns.nic.it	10720
it	IN	NS	a.dns.it	10720
…	…	NS	…	…
dns.nic.it	IN	A	192.12.192.5	10613
dns.nic.it	IN	AAAA	2a00:d40:1:1::5	10800
…	…	A	…	…
unipv.it	IN	NS	ipv36.unipv.it	300
unipv.it	IN	NS	ipv512.unipv.it	300
ipv36.unipv.it	IN	A	193.204.35.27	300
ipv36.unipv.it	IN	AAAA	2001:760:2000:2000::27	300
…	…	A	…	…

ALL Name Servers of `unipv.it`

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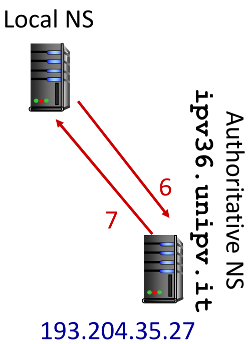
Cache content: final query



Name	Class	Type	Value	TTL
it	IN	NS	dns.nic.it	10720
it	IN	NS	a.dns.it	10720
…	…	NS	…	…
dns.nic.it	IN	A	192.12.192.5	10613
dns.nic.it	IN	AAAA	2a00:d40:1:1::5	10800
…	…	A	…	…
unipv.it	IN	NS	ipv36.unipv.it	300
unipv.it	IN	NS	ipv512.unipv.it	300
ipv36.unipv.it	IN	A	193.204.35.27	300
ipv36.unipv.it	IN	AAAA	2001:760:2000:2000::27	300
…	…	A	…	…
eva.unipv.it	IN	A	193.302.34.2	300

IP address of **eva.unipv.it**

Cache management

Caching is another source of loose coherence of the database: not every resolver is necessarily seeing the same content at a given instant of time

TTL measures the lifetime in the cache of each RR (i.e., how long a record can be stored in the cache before requiring a refresh)

As soon as a cached RR expires (i.e., TTL=0), the record is removed from the cache

The default TTL for the Resource Records of a domain (expressed in seconds) is stored in the SOA RR

Choice of the TTL is critical for the efficiency of the lookup phase

Basic rules [RFC 1912 “Common DNS Operational and Configuration Errors” published in 1996]

Choice of the TTL

“Small” TTL: RRs stored in the cache will expire quickly

+ Quick propagation of the updates
– Delays in the lookups

“Big” TTL: RRs stored in the cache will stay there for long time

+ Faster lookups
– Cache poisoning

General rules: “big” TTLs for Root and TLD Name Servers, “small” TTLs for Second level domain Name Servers

The default value of the TTL for the records of a domain is a configuration option of the domain administrator

Some examples of TTL choices

```
root-servers.net
        primary name server = a.root-servers.net
        responsible mail addr = nstld.verisign-grs.com
        serial  = 2026030500
        refresh = 14400 (4 hours)
        retry   = 7200 (2 hours)
        expire  = 1209600 (14 days)
        default TTL = 3600000 (41 days 16 hours)
```

```
net
        primary name server = a.gtld-servers.net
        responsible mail addr = nstld.verisign-grs.com
        serial  = 1774559662
        refresh = 1800 (30 mins)
        retry   = 900 (15 mins)
        expire  = 604800 (7 days)
        default TTL = 900 (15 mins)
```

```
it
        primary name server = dns.nic.it
        responsible mail addr = hostmaster.nic.it
        serial  = 2026032522
        refresh = 10800 (3 hours)
        retry   = 900 (15 mins)
        expire  = 604800 (7 days)
        default TTL = 3600 (1 hour)
```

```
ai
        primary name server = v0n0.nic.ai
        responsible mail addr = hostmaster.donuts.email
        serial  = 1774473321
        refresh = 7200 (2 hours)
        retry   = 900 (15 mins)
        expire  = 1209600 (14 days)
        default TTL = 3600 (1 hour)
```

```
io
        primary name server = a0.nic.io
        responsible mail addr = hostmaster.donuts.email
        serial  = 1774558819
        refresh = 7200 (2 hours)
        retry   = 900 (15 mins)
        expire  = 1209600 (14 days)
        default TTL = 3600 (1 hour)
```

And some others

```
unipv.it
        primary name server = ipv36.unipv.it
        responsible mail addr = root.ipv36.unipv.i
        serial  = 2026032603
        refresh = 12000 (3 hours 20 mins)
        retry   = 1200 (20 mins)
        expire  = 1209600 (14 days)
        default TTL = 300 (5 mins)
```

```
ucsd.edu
        primary name server = act-vnios-2.ucsd.edu
        responsible mail addr = hostmaster.ucsd.edu
        serial  = 1462920710
        refresh = 10800 (3 hours)
        retry   = 3600 (1 hour)
        expire  = 2419200 (28 days)
        default TTL = 900 (15 mins)
```

```
google.com
        primary name server = ns1.google.com
        responsible mail addr = dns-admin.google.com
        serial  = 889730864
        refresh = 900 (15 mins)
        retry   = 900 (15 mins)
        expire  = 1800 (30 mins)
        default TTL = 60 (1 min)
```

```
blog.google
        primary name server = ns1.googledomains.com
        responsible mail addr = dns-admin.google.com
        serial  = 2017070401
        refresh = 21600 (6 hours)
        retry   = 3600 (1 hour)
        expire  = 1209600 (14 days)
        default TTL = 300 (5 mins)
```

```
twitch.tv
        primary name server = ns-219.awsdns-27.com
        responsible mail addr = awsdns-hostmaster.amazon.com
        serial  = 1
        refresh = 300 (5 mins)
        retry   = 150 (2 mins 30 secs)
        expire  = 604800 (7 days)
        default TTL = 60 (1 min)
```

```
tiktok.com
        primary name server = a9-66.akam.net
        responsible mail addr = hostmaster.akamai.com
        serial  = 1554957832
        refresh = 43200 (12 hours)
        retry   = 7200 (2 hours)
        expire  = 604800 (7 days)
        default TTL = 7200 (2 hours)
```


Reverse mapping

DNS lookup provides the IP address(es) associated with a given hostname (*forward mapping*)

It is also useful to obtain the hostname associated with a given IP address (*reverse mapping*)

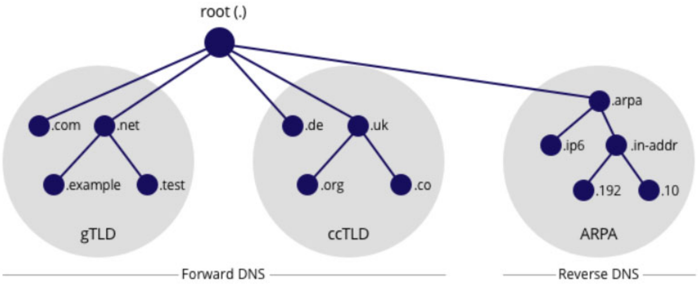
This reverse lookup is implemented by a separate parallel tree (keyed by IP address) under reserved domains, i.e., **`in-addr.arpa`** for IPv4 and **`ip6.arpa`** for IPV6

The TLD is **`arpa`** (Address and Routing Parameter Area)

The nodes of a reverse mapping zone are administered by the owner of the block of IP addresses

The reverse lookup process is based on the **`PTR`** resource record

Reverse mapping (cont'd)



[Source: https://www.ripe.net]

What is the hostname associated with **`192.58.128.30`**?

Query: **`30.128.58.192.in-addr.arpa. IN PTR`**

`30.128.58.192.in-addr.arpa. 3600 IN PTR j.root-servers.net.`

Answer: **`j.root-servers.net`**

Resource Record

Used by web analytics tools (e.g., visitor identification) and for mail services (e.g., for fighting spam)